EHR-QC

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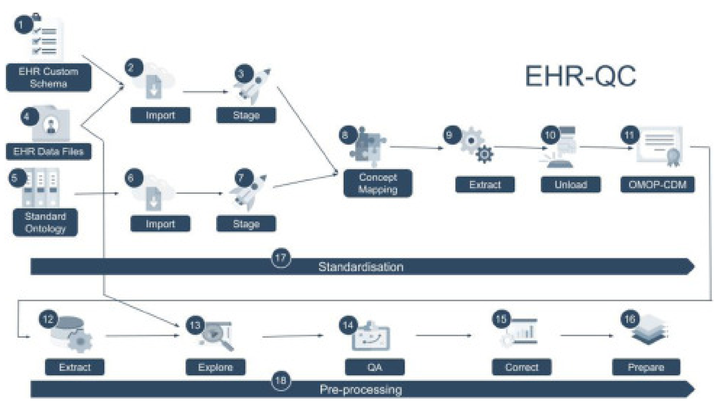
INTRODUCTION

A streamlined pipeline for automated electronic health records (EHR) standardisation and preprocessing to predict clinical outcomes



Challenges:

1. Lack of standardisation in EHR data
   a. Standard schema/structure
   b. Standard vocabulary
2. The presence of anomalies, such as missing data, outliers, errors, and inconsistencies

Tool	Input	Schema standardisation	QC		Concept mapping	Anomaly correction	Data wrangling
			Summary	Individual			
Achilles[b]	Restricted	N	Y	N	None	N	N
Data Quality Dashboard (DQD)[b]	Restricted	N	Y	Y	None	N	N
ARES[b]	Restricted	N	Y	Y	None	N	N
White Rabbit[b]	Flexible	N	Y	N	None	N	N
Rabbit In A Hat[b]	Flexible	Y[a]	N	N	None	N	N
Usagi[b]	Flexible	N	N	N	Manual	N	N
MIRACUM DQA	Restricted	N	Y	Y	None	N	N
MosaicQA	Restricted	N	Y	Y	None	N	N
Extract, transform, load (ETL) framework	Flexible	Y	N	N	None	N	N
EHR-QC	**Flexible**	**Y**	**Y**	**Y**	**Automatic**	**Y**	**Y**

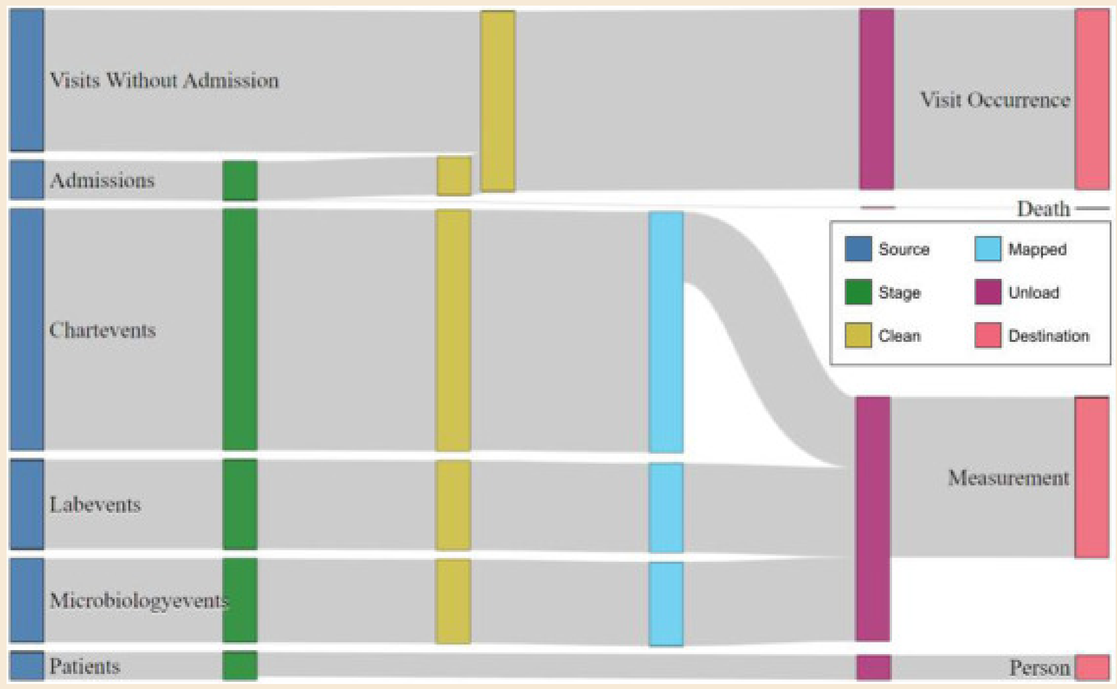
a It generates documentation, but not source code for the ETL process.

b These tools all operate independent to each other and part of a single workflow developed by the OHDSI consortium.

CONCLUSION

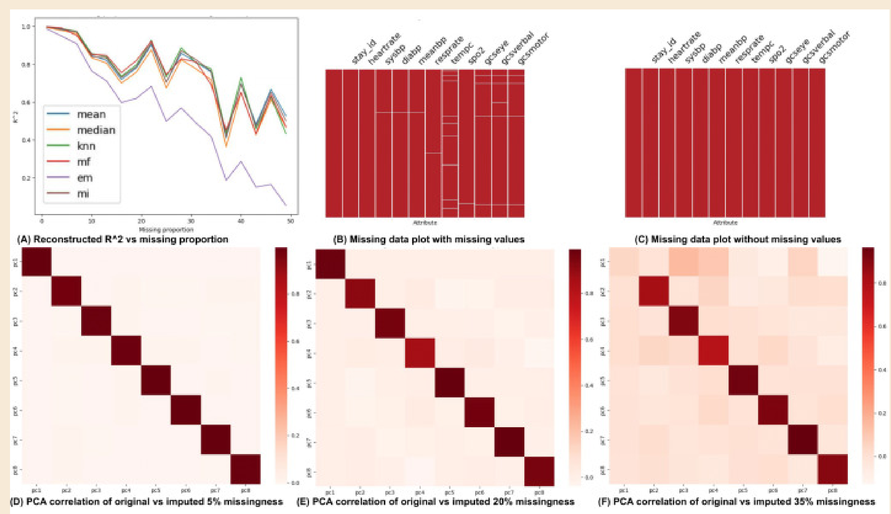
- EHR-QC offers a comprehensive and user-friendly resolution for managing healthcare data, ensuring research outcomes that are efficient, reproducible, and robust
- We hope this will contribute to the wider acceptance of EHR standards and facilitate the creation of more versatile data-driven models
- We have released the source code to the public, using a permissive MIT licence to support open research and community participation at
- A complete step-by-step guide to running the pipeline is available in the documentation

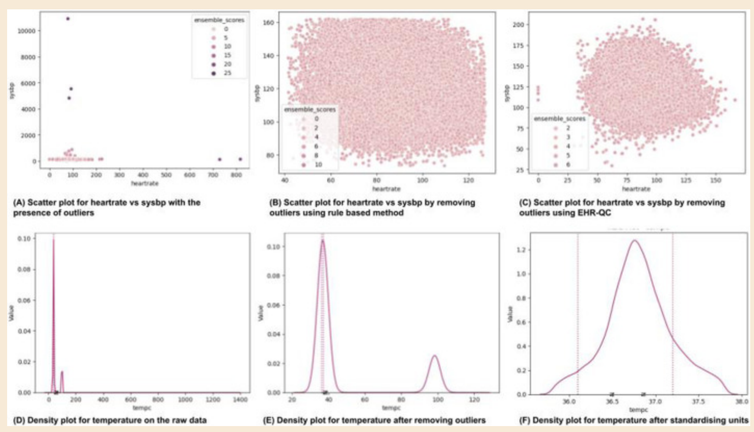
RESULTS



The figure illustrates the data flow diagram of four different entities. In all cases, the data from the source schema is imported into a staging table which is transformed into standard entities after performing cleaning and mapping to a desired standard. The standard entities are dumped in the unloading table which will be finally pushed to the destination OMOP-CDM schema.

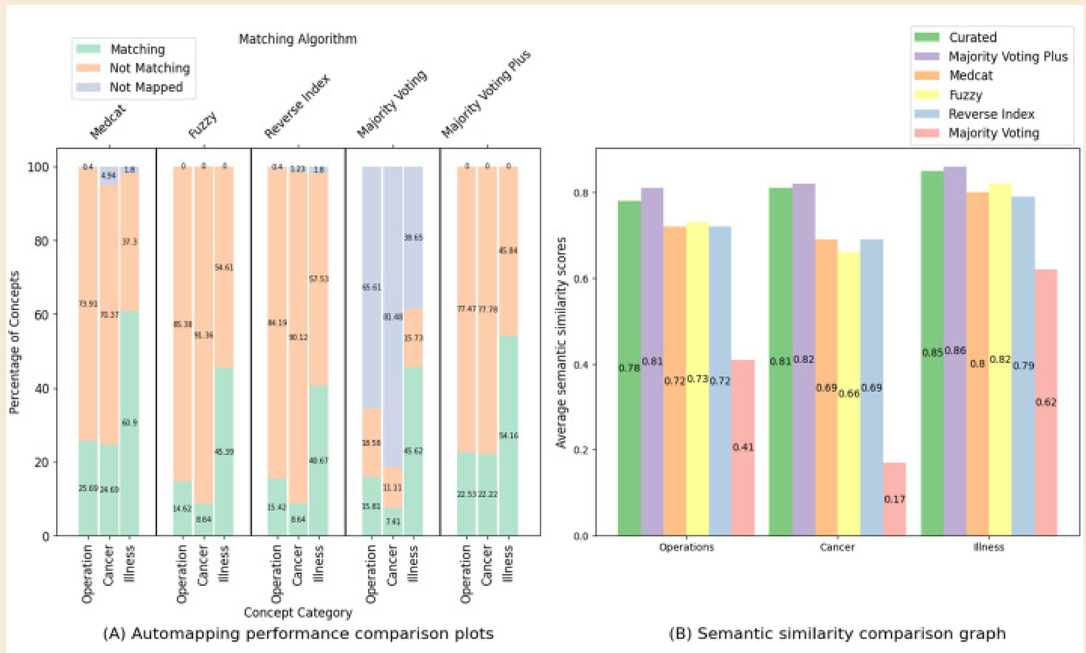
- Plot A, the reconstruction r-squared score plotted against missing proportions, spanning from 0 to 50 for various imputation techniques.
- Plots B and C display two of the EHR-QC exploration plots, illustrating data with and without missing values.
- Plots D, E, and F display the correlation between the principal components of the original data and the imputed data for simulated missingness of 5%, 20%, and 35%, respectively.





- Plots A, B, and C showcase the adaptive outlier detection capability of EHR-QC. Plot A presents a scatter plot of the data with outliers, while Plot B is derived from the same dataset after applying a conventional univariate rule-based method for outlier removal. Plot C, on the other hand, is obtained by applying an unsupervised outlier removal method.
- Plots 5D, 5E, and 5F demonstrate how EHR-QC plots help address data inconsistencies, using the temperature attribute as an example

- Plot A is an illustration of the percentage of coverage and mapping performance for Fuzzy, Reverse Index, Medcat, Majority Voting, and Majority Voting Plus techniques under three concept categories.
- In the plot B, the average "Semantic Similarity Score" is plotted for the expert-curated concepts along with the scores for the mappings from Fuzzy, Reverse Index, Medcat, Majority Voting, and Majority Voting Plus techniques.



Acknowledgements: We acknowledge contributions from Esha Singh and Geeta Kole for their initial analysis of MIMIC to OMOP conversion workflow. Additionally, we extend our thanks to Jerico Revote from Monash eResearch Centre for his invaluable support throughout the research process. The authors also extend their sincere appreciation to the open research community responsible for making the resources available which were instrumental in facilitating the execution of this research.

Cite: Ramakrishnaiah, Y. *et al.* (2023) 'EHR-QC: A streamlined pipeline for Automated Electronic Health Records standardisation and preprocessing to predict clinical outcomes', *Journal of Biomedical Informatics*, 147, p. 104509. doi:10.1016/j.jbi.2023.104509.

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